

4.1 Forces: Turning Effects & Equilibrium

Question Paper

Course	CIEA Level Physics
Section	4. Forces, Density & Pressure
Topic	4.1 Forces: Turning Effects & Equilibrium
Difficulty	Easy

Time allowed: 40

Score: /30

Percentage: /100

Question 1a

Define the following

(i)
moment of a force

[2]

(ii)
centre of gravity.

[1]

[3 marks]

Question 1b

A teacher sets some Physics students the task of finding the weight of a broom using only a ruler, some thin, light string, and a set of laboratory masses. The equipment is initially arranged as shown in Fig. 1.1.

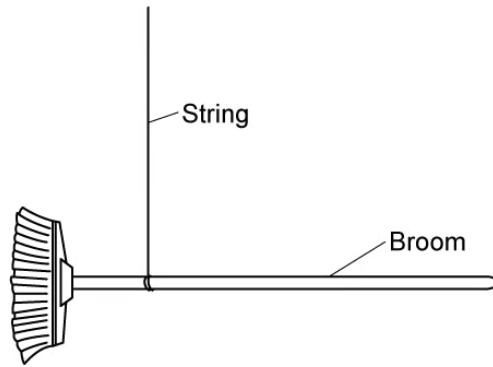


Fig. 1.1.

The students are not allowed to use scales.

For the arrangement in Fig. 1.1.

- (i)
State why the broom is hanging horizontally.

[2]

- (ii)
State what measurement the students should make to start their investigation to find the weight of the broom.

[1]

[3 marks]

Question 1c

The measurement to the position of the string found in part (b) is shown in Fig. 1.2.

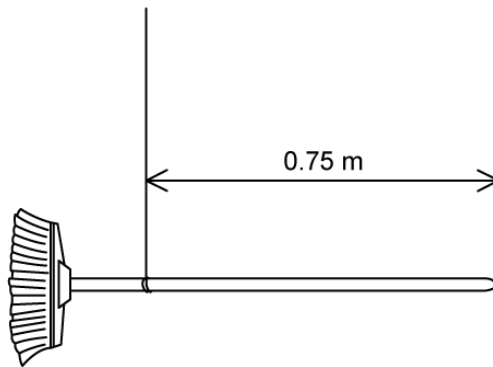


Fig. 1.2.

Distance from string to the end of the handle = 0.75 m

The string is moved towards the end of the handle, so that the broom is in equilibrium when a 0.5 kg mass is hung a distance of 0.45 m from the string as shown in Fig. 1.3.

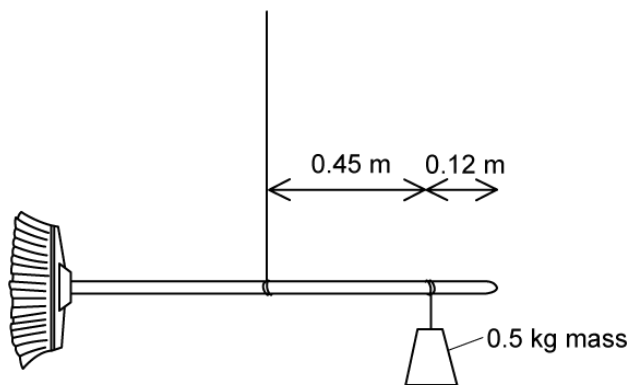


Fig. 1.3

Determine

(i)
the distance between the centre of mass of the broom and the point where the string is attached.

[1]

(ii)
the force exerted by the 0.5 kg mass.

[1]

[2 marks]

Question 1d

Calculate the weight of the broom.

[2 marks]

Question 2a

A metre rule is balanced at its on a pivot at its mid-point.

A force of 10.0 N acts at a distance of 8.0 cm from one end of the rule as shown in Fig. 1.1.

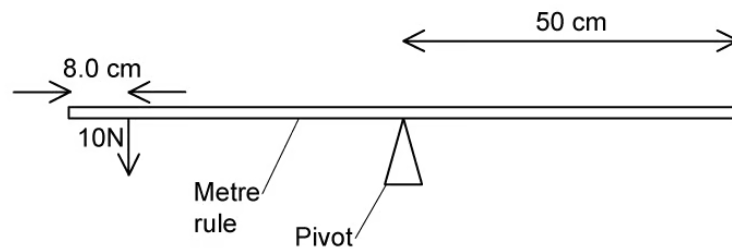


Fig. 1.1

Calculate the moment of the 10.0 N force about the pivot.

moment = Nm

[3 marks]

Question 2b

A mass is placed at a point 35 cm from the pivot, which makes the metre rule balance.
On Fig. 1.1, draw an arrow to show the position of the mass and the direction of the force it exerts.

[2 marks]**Question 2c**

Calculate the **mass** which has been placed to balance the ruler in part (a).

[4 marks]**Question 3a**

For a couple

(i)

State the definition

[1]

(ii)

State the conditions which have to be true for the forces in a couple.

[3]**[4 marks]**

Question 3b

Fig. 1.1 shows a pair of forces acting on a rotating wheel.

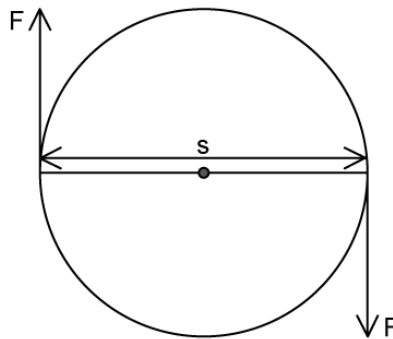


Fig. 1.1

(i)

State the definition and units of torque

[1]

(ii)

For the couple in Fig. 1.1 state the equation which can be used to determine the torque.

[1]

[2 marks]

Question 3c

For the couple in Fig. 1.1. the two forces, F are both 15 N and the perpendicular distance, s , between them is 0.5 m.

Calculate the torque of this couple.

[2 marks]

Question 3d

The forces in Fig. 1.1. are changed so that they act at an angle of 60° to the horizontal distance between them as shown in Fig. 1.2.

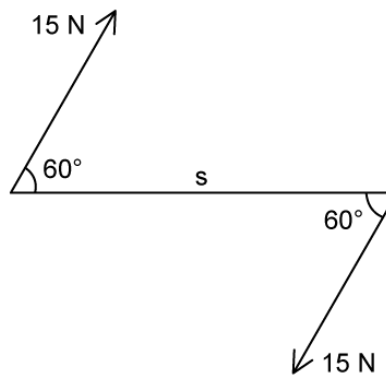


Fig. 1.2.

Calculate the new torque on the wheel.

[3 marks]

Model Answers: Easy

1a

(a)

(i) A moment of a force is defined as:

- The turning effect of a force; [1 mark]
- Moment = Force $\times$ perpendicular distance from the pivot **OR** $M = Fd$; [1 mark]

(ii) Centre of gravity is defined as:

- The single point where the weight (of a body) can be taken as acting from; [1 mark]

[Total: 3 marks]

1b

(b) For the arrangement in Fig. 1.1.

(i) State why the broom is hanging horizontally:

- The broom is in equilibrium; [1 mark]
- The string is fixed at the centre of mass; [1 mark]

(ii) State what measurement the students should make to start their investigation to find the weight of the broom:

- Distance from either end to the string; [1 mark]

[Total: 3 marks]

The students will need to know the position of the centre of mass, so they definitely need the distance from one end to the string, which must be at the centre of mass for

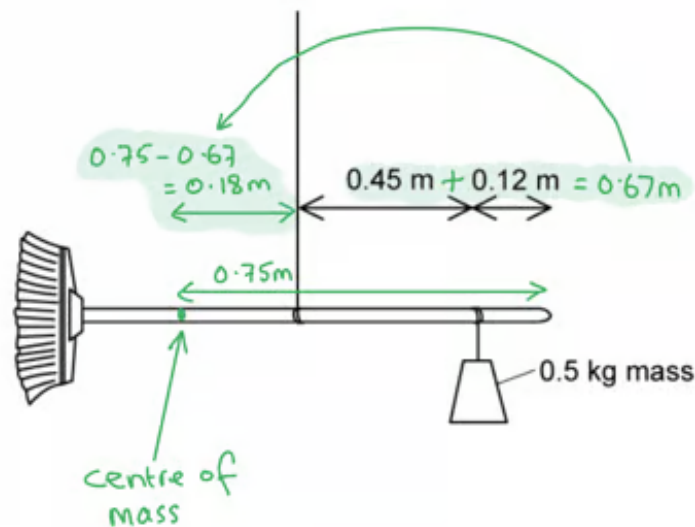
equilibrium.

1c

(c)

(i) Determine the distance between the centre of mass of the broom and the point where the string is attached:

- Total distance from end of handle to centre of mass = 0.75 m



- Distance to centre of mass, $d = 0.75 - (0.45 + 0.12) = 0.18 \text{ m}$; [1 mark]

(ii) Determine the force exerted by the 0.5 kg mass:

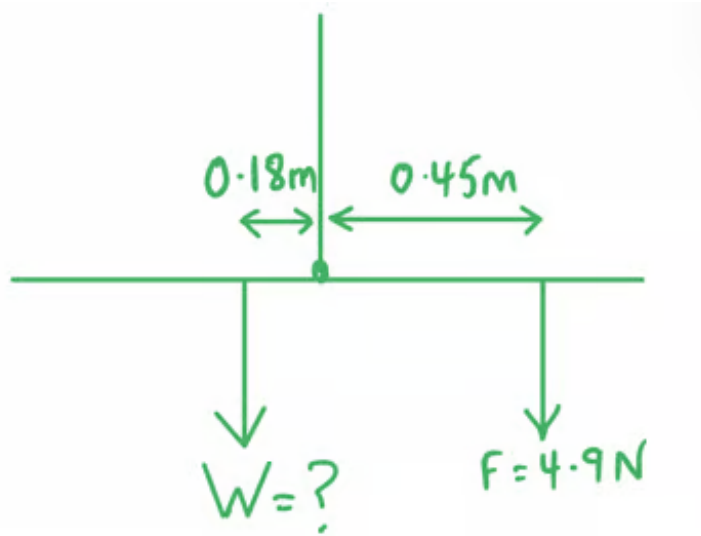
- force exerted = weight acting downwards
- $W = mg = 0.5 \times 9.81 = 4.9 \text{ N}$ [1 mark]

[Total: 2 marks]

1d

(d) Calculate the weight of the broom:

Sketch a diagram to show the forces and distances:



Apply the principle of moments to the point where the string is attached to the broom:

- Sum of anti-clockwise moments = sum of clockwise moments
- $W \times 0.18 = 4.9 \times 0.45$ [1 mark]
- $W = \frac{4.9 \times 0.45}{0.18} = 12.25$
- $W = 12 \text{ N (2 s.f.)}$ [1 mark]

[Total: 2 marks]

Many students are reluctant to 'waste time' sketching diagrams which don't get marks. This is not a waste of time!

The diagram ensures you write the correct values in your calculation.

Although not marked directly, it is an essential part of the process to find the solution.

2a

(a)

To calculate the moment:

List the known quantities:

- Force, $F = 10 \text{ N}$

- Distance of force from one end of pivot = 8.0 cm
- Therefore distance of force from pivot, $d = 50 - 8 = 42 \text{ cm}$ **or** 0.42 m [1 mark]

Recall the equation for a moment and calculate:

- $M = Fd$
- $M = 10 \times 0.42$ [1 mark]
- $M = 4.2 \text{ Nm}$ [1 mark]

[Total: 3 marks]

In a question like this one the first mark is for identifying the correct distance, so at that stage you could answer in cm or m and still get the mark.

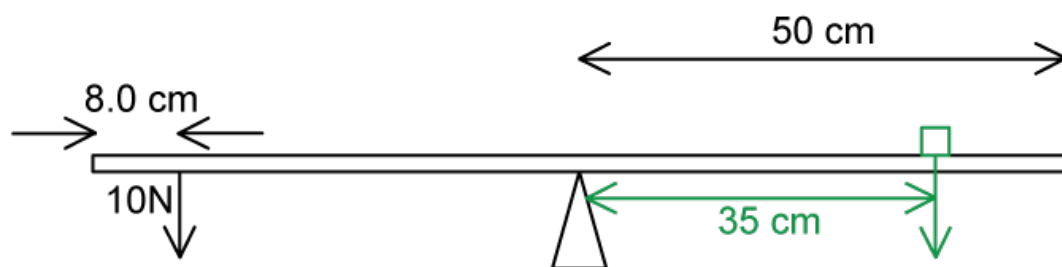
However the question specifies **Nm** as the final unit, so to get the next two marks you have to convert to **metres**.

Keep an eye on units in moments questions as they can be tricky.

2b

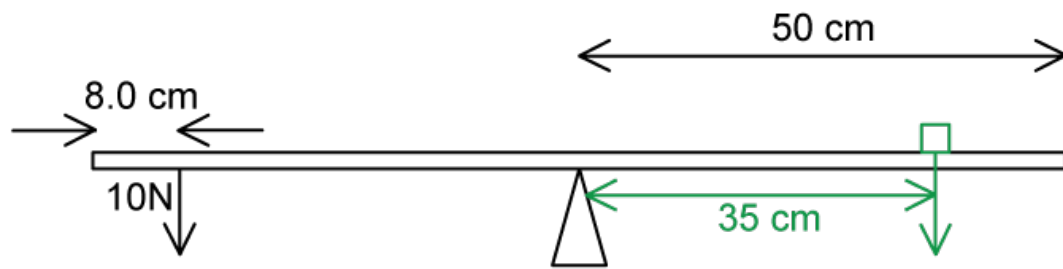
(b) On Fig. 1.1, draw an arrow to show the position of the mass and the direction of the force it exerts.

A correctly annotated diagram looks like:



- Downwards arrow marked on the right of the pivot (so it causes a clockwise moment); [1 mark]
- Mass / force arrow 35 cm from pivot (distance labelled or by eye); [1 mark]

[Total: 2 marks]



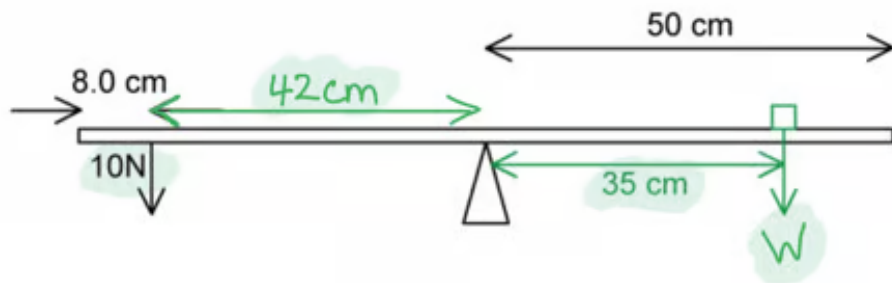
- Downwards arrow marked on the right of the pivot (so it causes a clockwise moment); [1 mark]
- Mass / force arrow 35 cm from pivot (distance labelled or by eye); [1 mark]

[Total: 2 marks]

2c

(c) Calculate the **mass** which has been placed to balance the ruler in part (a):

Use the diagram and distances from parts (a) and (b):



Apply the principle of moments since the ruler is in equilibrium:

- sum of clockwise moments = sum of anti-clockwise moments
- $W \times 35 = 10 \times 42$ [1 mark]
- $W = \frac{420}{35} = 12 \text{ N}$ [1 mark]

Convert weight to mass:

- $m = \frac{W}{g} = \frac{12}{9.81}$ [1 mark]
- $m = 1.2 \text{ kg}$ [1 mark]

[Total: 4 marks]

3a

(a) For a couple

(i) State the definition:

- A pair of forces that act to produce rotation only; [1 mark]

ii) State the conditions which have to be true for the forces in a couple:

The forces are:

- Equal in magnitude; [1 mark]
- Opposite in direction; [1 mark]
- Perpendicular to the distance between them; [1 mark]

[Total: 4 marks]

3b

(b)

(i) State the definition and units of torque:

- The moment of a couple **AND** Nm; [1 mark]

(ii) For the couple in Fig. 1.1 state the equation which can be used to determine the torque:

- Torque / T / $\tau = Fs$; [1 mark]

[Total: 2 marks]

3c

(c) Calculate the torque of this couple:

List the known quantities:

- Force, $F = 15 \text{ N}$
- Perpendicular distance between forces, $s = 0.5 \text{ m}$

Write the equation and calculate:

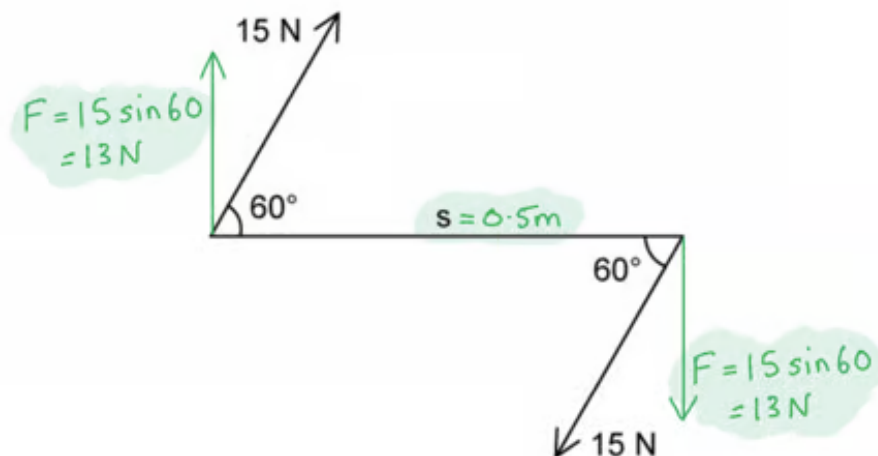
- $\tau = F \times s = 15 \times 0.5$ [1 mark]
- $\tau = 7.5 \text{ Nm}$ [1 mark]

[Total: 2 marks]

3d

(d) Calculate the new torque on the wheel.

Determine the component of the force, F which is perpendicular to the horizontal line between the forces:



- $F = 15 \times \sin 60 = 12.99$ **or** 13 N [1 mark]

Calculate the torque:

- $\tau = F \times s = 12.99 \times 0.5$ **or** 13×0.5 [1 mark]
- $\tau = 6.5 \text{ N (2 s.f.)}$ [1 mark]

[Total: 3 marks]

